

Lab 08: Inter-VLAN Routing (Router-on-a-Stick)

Enabling Communication Between VLANs Using Sub-interfaces

Lab #8

Topic: VLANs &
Inter-VLAN Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build on VLAN segmentation by enabling routing between VLANs.
- Configure sub-interfaces on a router using 802.1Q encapsulation (Router-on-a-Stick).
- Configure a trunk port on the switch to carry all VLANs to the router.
- Verify that PCs in different VLANs can now ping each other via the router.
- Relate to Network+ objective: 1.4 – Explain routing concepts including inter-VLAN routing.

2. Network Topology

Same switch as Lab 07 (VLANs 10, 20, 30), but now a router is connected via a single trunk link. The router uses sub-interfaces — one per VLAN — to route between them.

Sub-interface	VLAN	Encapsulation	IP (Gateway)	Subnet
Fa0/0.10	VLAN 10 (SALES)	dot1Q 10	192.168.10.1	192.168.10.0/24
Fa0/0.20	VLAN 20 (TECH)	dot1Q 20	192.168.20.1	192.168.20.0/24
Fa0/0.30	VLAN 30 (MGMT)	dot1Q 30	192.168.30.1	192.168.30.0/24

3. Step-by-Step Configuration

3.1 Switch – Configure Trunk Port to Router

```
Switch(config)# interface FastEthernet0/24    ! Port connected to router
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch# write memory
```

3.2 Router – Sub-interfaces with 802.1Q

```
Router(config)# interface FastEthernet0/0
Router(config-if)# no shutdown
```

```

Router(config-if)# exit

Router(config)# interface FastEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface FastEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit

Router(config)# interface FastEthernet0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
Router(config)# exit
Router# write memory

```

3.3 Update PC Default Gateways

VLAN 10 PCs – GW	192.168.10.1
VLAN 20 PCs – GW	192.168.20.1
VLAN 30 PCs – GW	192.168.30.1

4. Verification

```

PC1 (VLAN10)> ping 192.168.20.10    ! VLAN10 → VLAN20 via router
PC1 (VLAN10)> ping 192.168.30.10    ! VLAN10 → VLAN30 via router
Router# show ip route                ! Should see all 3 connected subnets
Switch# show interfaces trunk        ! Fa0/24 should show all VLANs allowed

```

5. Key Takeaways

- Router-on-a-Stick uses ONE physical link with 802.1Q sub-interfaces to route between multiple VLANs.
- The physical parent interface must be up (no shutdown) for sub-interfaces to work.
- Trunk ports send and receive frames tagged with a VLAN ID; access ports are untagged.
- Each sub-interface acts as the default gateway for its respective VLAN.
- A Layer 3 switch is a more scalable alternative — it performs inter-VLAN routing in hardware.